

Data Science Final Project
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4371ORRN

Topic

Amazon Product Reviews – Multi-Class Sentiment Analysis and Topic Modelling.

Summary of Topic & Background

E-commerce sites get tons of customer reviews which offer great insights on how products perform and what customers think. Yet, going through all that text manually is impossible. That's where this project steps in – it uses machine learning and natural language processing to figure out if reviews are positive or negative, identify primary talking points, and see if reviews are relevant.

This helps businesses big time as they can use the info to boost their product quality, solve common problems and ultimately make their customers happier.

Research Questions

How effective can machine learning and NLP techniques extract customer sentiment, identify key discussion topics, and assess the quality and authenticity of Amazon product reviews to generate actionable insights for e-commerce decision-making?

Project Objectives

To create a multi-signal fake review filter, we use the verified purchase status, make sure the review matches the star rating, check the linguistic patterns, and analyze reviewer behavior to pick out genuine reviews beforehand. We then build several sentiment classifiers like Logistic Regression, Random Forest, and DistilBERT, comparing their accuracy on real reviews, aiming for at least 85% correct in star rating predictions on Amazon.

Next, we find the main topics customers discuss by using both LDA and BERTopic. This reveals top themes such as product quality, delivery troubles, price worries, and how easy or difficult the product is to use. We figure out what drives both happiness and frustration based on these.

After that, we come up with a review quality score. This score combines factors like review length, strength of sentiment, number of helpful votes, and those authenticity checks to decide which reviews matter.

Lastly, we dig into and draw pictures showing how authentic reviews are linked to overall sentiment, the big discussion topics, and the quality of the reviews themselves. This helps e-commerce businesses get useful info to guide their decisions.

Reference List

Breiman, L. (2001). Random Forests. Machine Learning, 45(1), 5–32.
<https://doi.org/10.1023/A:1010933404324>

Devlin, J., Chang, M. W., Lee, K., & Toutanova, K. (2019). BERT: Pre-training of deep bidirectional transformers for language understanding. In Proceedings of NAACL-HLT 2019 (pp. 4171–4186). <https://arxiv.org/abs/1810.04805>

Grootendorst, M. (2022). BERTopic: Neural topic modeling with a class-based TF-IDF procedure. arXiv Preprint arXiv:2203.05794. <https://arxiv.org/abs/2203.05794>

Jain, P. K., Saravanan, V., & Pamula, R. Sentiment Analysis Based on Deep Learning: A Comparative Study. Electronics, 9(3), 483. <https://doi.org/10.3390/electronics9030483>

TimeLine

Task	Timeline	Description
Literature Review	Week 1	Review NLP, sentiment analysis, and topic modeling literature.
Data Download & EDA	Week 2	Download Kaggle dataset, explore structure, analyze class distribution.
Data Preprocessing	Week 3	Clean data, tokenize, lemmatize, handle missing values.
Feature Engineering	Week 4	Create TF-IDF, word embeddings, derived features.
Model Development	Week 5	Train sentiment classifiers and compare performance.
Topic Modeling	Week 6	Apply LDA and BERTopic to identify themes.
Analysis & Report	Week 7	Correlate findings and write comprehensive report. SUBMISSION DUE.

Task / Week	May 25 - May 31	Jun 01 - Jun 07	Jun 08 - Jun 14	Jun 15 - Jun 21	Jun 22 - Jun 28	Jun 29 - Jul 05	Jul 06 - Jul 12	Jul 13 - Jul 19
Literature Review	✓	✓						
Data Download & EDA		✓	✓					
Data Preprocessing			✓	✓				
Feature Engineering				✓	✓			
Model Development					✓	✓	✓	
Topic Modeling							✓	✓
Analysis & Report								✓
Assessments		✓						✓

Data Management Plan

The project uses the Amazon US Customer Reviews Dataset from Kaggle, which has reviews, star ratings, titles, and helpful vote counts. This dataset captures what U.S. customers think about various products. We'll download it in CSV form from Kaggle and use Python to do sentiment analysis and topic modelling on it.

Dataset Link:

<https://www.kaggle.com/datasets/cynthiarempel/amazon-us-customer-reviews-dataset>.

Metadata

The data set was downloaded from Kaggle as a ZIP file, extracted as TSV format and converted to CSV format for ease of processing using Python. The dataset consists of Amazon customer reviews, star ratings, review titles, helpful votes and product information. It consists of millions of review records (around 130 million records depending on the selected categories). The dataset used in the project contains approximately 6.9 million datasets. Data files are in CSV format, code files are in Python (.py) scripts, and Google Collab Notebooks (.ipynb). The estimated storage need is 3-5 GB for raw data and processed datasets and trained models.

Version Control

We will use GitHub for version control and project management. The GitHub link is as follows: <https://github.com/revathisankar44-SAU/SAUI/Final-Project>. Code will be committed on a weekly basis, with additional commits for major project milestones. The naming conventions of the files format will be YYYY-MM-DD_task_v#: The repository will have separate folders for raw data, processed data, notebooks, models, and reports.

README File

The GitHub repository will include a README file that will include the overview of the project, description of the dataset, software requirements, installation instructions, project structure, usage guidelines, and a summary of results.

Storage & Security

Project files will be stored on GitHub and backed up weekly to OneDrive. The GitHub repository will be shared with supervisor and the markers. The dataset contains no personally identifiable information and therefore meets ethical and data security requirements.

Ethics

1. Compliance with GDPR

The dataset does not contain any personally identifiable information (PII) such as names, addresses, email address or contact details. The project therefore poses a minimal risk in terms of GDPR and does not involve the processing of personal data.

2. UH Ethical Policies

The project complies with the University ethical guidelines for data science research. The study uses secondary data that is publicly available, and does not involve human participants, surveys, interviews or experiments.

3. Data Use Permission

The data set is publicly available in the public domain via Kaggle for research and educational purposes. The project will adhere to the licensing and usage terms of the dataset.

4. Ethical Data Collection

Amazon got these reviews from people who used their website to buy things. They wanted to know what customers thought about the products and services they sold. The information they shared with us only has reviews and no private information about the people who wrote them. It seems like Amazon did not do anything when they collected this data. They just used the reviews to get feedback from customers, on the products and services.